TUTORIAL
CHAPTER 1: UNIT, MEASUREMENT AND MATH REVIEW

Problems

2. **a.** Suppose the displacement of an object is related to time according to the expression $x = Bt^2$. What are the dimensions of B?

b. A displacement is related to time as $x = A \sin(2\pi ft)$, where A and f are constants. Find the dimensions of A. *Hint:* A trigonometric function appearing in an equation must be dimensionless.

3. S A shape that covers an area A and has a uniform height h has a volume $V = Ah$.

a. Show that $V = Ah$ is dimensionally correct.

b. Show that the volumes of a cylinder and of a rectangular box can be written in the form $V = Ah$, identifying A in each case. (Note that A, sometimes called the "footprint" of the object, can have any shape and that the height can, in general, be replaced by the average thickness of the object.)

5. Newton's law of universal gravitation is represented by

$$F = G\frac{Mm}{r^2}$$

where F is the gravitational force, M and m are masses, and r is a length. Force has the SI units $\mathrm{kg \cdot m/s^2}$. What are the SI units of the proportionality constant G?

9. A fathom is a unit of length, usually reserved for measuring the depth of water. A fathom is approximately 6 ft in length. Take the distance from Earth to the Moon to be $250\,000$ miles, and use the given approximation to find the distance in fathoms.

10. A small turtle moves at a speed of 186 furlongs per fortnight. Find the speed of the turtle in centimeters per second. Note that $1\ \text{furlong} = 220\ \text{yards}$ and $1\ \text{fortnight} = 14\ \text{days}$.

16. V BIO Suppose your hair grows at the rate of $1/32$ inch per day. Find the rate at which it grows in nanometers per second. Because the distance between atoms in a molecule is on the order of 0.1 nm, your answer suggests how rapidly atoms are assembled in this protein synthesis.

18. T A house is 50.0 ft long and 26 ft wide and has 8.0-ft-high ceilings. What is the volume of the interior of the house in cubic meters and in cubic centimeters?

19. The amount of water in reservoirs is often measured in acre-ft. One acre-ft is a volume that covers an area of one acre to a depth of one foot. An acre is $43\ 560\ \text{ft}^2$. Find the volume in SI units of a reservoir containing 25.0 acre-ft of water.

23. A rectangular airstrip measures 32.30 m by 210 m, with the width measured more accurately than the length. Find the area, taking into account significant figures.

24. Use the rules for significant figures to find the answer to the addition problem $21.4 + 15 + 17.17 + 4.003$.

25. V A carpet is to be installed in a room of length 9.72 m and width 5.3 m. Find the area of the room, retaining the proper number of significant figures.

28. The speed of light is now defined to be $2.997\ 924\ 58 \times 10^8 \text{m/s}$. Express the speed of light to

a. three significant figures,

b. five significant figures, and

c. seven significant figures.

29. A rectangle has a length of (2.0 ± 0.2) m and a width of (1.5 ± 0.1) m. Calculate

a. the area and

b. the perimeter of the rectangle, and give the uncertainty in each value.

30. The radius of a circle is measured to be (10.5 ± 0.2) m. Calculate

a. the area and

b. the circumference of the circle, and give the uncertainty in each value.

36. BIO Treat a cell in a human as a sphere of radius $1.0\ \mu\mathrm{m}$.

a. Determine the volume of a cell.

b. Estimate the volume of your body.

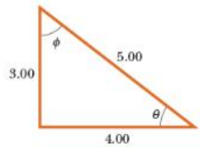
c. Estimate the number of cells in your body.

39. T A point is located in a polar coordinate system by the coordinates $r = 2.5\ \mathrm{m}$ and $\theta = 35^\circ$. Find the x- and y-coordinates of this point, assuming that the two coordinate systems have the same origin.

40. A certain corner of a room is selected as the origin of a rectangular coordinate system. If a fly is crawling on an adjacent wall at a point having coordinates $(2.0, 1.0)$, where the units are meters, what is the distance of the fly from the corner of the room?

42. V Two points in a rectangular coordinate system have the coordinates $(5.0, 3.0)$ and $(-3.0, 4.0)$, where the units are centimeters. Determine the distance between these points.

Figure P1.49



49. In Figure P1.49, find

a. the side opposite θ,

b. the side adjacent to φ,

c. $\cos\theta$,

d. $\sin\varphi$, and

e. $\tan\varphi$.

56. QC An airplane flies 2.00×10^2 km due west from city A to city B and then 3.00×10^2 km in the direction of 30.0° north of west from city B to city C.

a. In straight-line distance, how far is city C from city A?

b. Relative to city A, in what direction is city C?

c. Why is the answer only approximately correct?

58. A force $\vec{\mathbf{F}}_1$ of magnitude 6.00 units acts on an object at the origin in a direction $\theta = 30.0^\circ$ above the positive x-axis (Fig. P1.58). A second force $\vec{\mathbf{F}}_2$ of magnitude 5.00 units acts on the object in the direction of the positive y-axis. Find graphically the magnitude and direction of the resultant force $\vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2$.

Figure P1.58